

Yaqin Hasan

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Open to relocation · Visa sponsorship required

Experience

Cambridge Kinetics (May 2024 - January 2026)

Software Engineer (Full Stack) • C# .NET, TypeScript, Azure, MongoDB, Vue, GraphQL

- Reduced customer onboarding configuration from hours to seconds by leading end-to-end delivery of LLM-powered workflow automation, owning requirements scoping, system architecture and team coordination.
- Aggregated customer data from multiple external systems by architecting and implementing a data synchronisation service, processing thousands of daily updates across heterogeneous schemas with minimised API overhead.
- Improved legacy REST API performance by implementing type-safe bulk operations, reducing 100 round-trips into a single call and supporting operations on 10k+ records.
- Increased product revenue by co-designing the commercial model for usage-based pricing and subscription add-ons, then extending the existing Stripe billing integration end-to-end without disrupting live subscriptions.
- Unified customer financial workflows by delivering Xero and QuickBooks integrations, implementing OAuth account linking, webhook ingestion and error reconciliation.
- Cut MongoDB Atlas costs by ~1000£/month by profiling query patterns, identifying excessive search memory consumption from read-heavy workloads, and migrating to dedicated Atlas Search nodes.
- Halved CI/CD pipeline run times by transitioning to hybrid local and cloud runners, and managed Azure production deployments.
- Onboarded and mentored engineers and SDETs from zero experience to full team contribution, covering SDLC principles, full stack development best practices and developer tooling.

Education

University of Southampton (2020 - 2023)

BSc Computer Science (Honours)

Specialist Modules: Advanced Computer Architecture & Systems, Real-time Computing & Embedded Systems (RTOS), Programming Language Concepts (compiler theory), Web Architecture & Hypertext Technologies (REST Theory)

Personal Projects

CryptIt • Desktop app

Cryptographically secure secret splitting & reconstruction desktop app built with Rust & Tauri.

Rust • Tauri • Vue • Cryptography • Desktop App Development

TrayRunner • macOS app

Spotlight-inspired menu bar utility for launching & scheduling local script-based workflows, built with Swift & SwiftUI.

Swift • SwiftUI • macOS Development • Desktop App Development • Shell Scripting

opsona • npm library

Open source TypeScript framework for building graph RAG pipelines using Fandom wiki data. Powering persona-based chat on mikoshibd.net, running fully client-side with WebLLM

TypeScript • AI • LLM • RAG • WebLLM • npm • Developer Tooling

opl-py • PyPI package

Open source Python SDK and dockerised web app for querying and analysing OpenPowerlifting data, with performance prediction using machine learning

Python • FastAPI • React • Docker • scikit-learn • PyTorch • PyPI • Machine Learning • Data analytics

STQL • Interpreter

Interpreter for a custom SQL-inspired RDF querying language, implemented in Haskell

Haskell • Interpreter Design • Compiler Theory • Formal Language Theory

ThymPi • Research project

Proof-of-concept robotics platform using sensor data to augment visual navigation systems for autonomously navigating robots.

Python • OpenCV • Machine Vision • Robotics • Autonomous Navigation • Final year project

Skills

Languages: C#, TypeScript, JavaScript, Python, Rust, Swift, SQL

Frontend: Vue, React, Vite, Pinia, SPA Development, Responsive UI

Backend: .NET, Node.js, FastAPI, GraphQL, REST APIs, Microservices, OAuth, Webhooks

Data: MongoDB, PostgreSQL, Database Design, Data Synchronisation, Schema Design

Infrastructure & DevOps: Azure, AWS, Docker, GitHub Actions, CI/CD Pipelines, Cloud Infrastructure

Testing & Practices: Vitest, XUnit, Playwright, TDD, Agile, Code Review